

Model Predictive Control

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github.com/silvasta/summary-mpc



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Requirements and Steps to MPC

- 1 Model of the System dynamics to state space
- 2 State Estimator track trajectory and disturbance
- 3 Optimal Control Problem define control strategy
- 4 Optimization problem mathematical formulation
- 5 Get Optimal Control Sequence solve op in time
- 6 Verify Closed-Loop Performance iterative tests

1 Nominal MPC

MPC Mathematical Formulation

$$\operatorname{argmin}_U \sum_{i=0}^{N-1} I(x_i, u_i) + I_f(x_N) \quad (1)$$

Constraints $x_0 = x(k)$
 $x_{i+1} = f(x_i, u_i)$ linear case : $Ax_i + Bu_i$
 $x_i \in \mathcal{X}$
 $u_i \in \mathcal{U}$
 $x_N \in \mathcal{X}_f$
 $I_f(\cdot)$ and $\mathcal{X}_f$ are chosen to mimic an infinite horizon.

What can go wrong with *standard* MPC?

- No feasibility guarantee, the problem may not have a solution
- No stability guarantee, trajectories may not converge to origin

Finite-horizon MPC may not satisfy constraints for all time!

Finite-horizon MPC may not be stable!

- An infinite-horizon provides stability and invariance.
- Infinite-horizon *faked* by forcing final state into an invariant set for which there exists invariance-inducing controller, whose infinite-horizon cost can be expressed in closed-form.

Stability of MPC - Main Result

Assumptions

- 1 Stage cost is strictly positive and only zero at the origin
- 2 Terminal set is **invariant** under local control law $\kappa_f(x_i)$:

$$x_{i+1} = Ax_i + B\kappa_f(x_i) \in \mathcal{X}_f \quad \forall x_i \in \mathcal{X}_f$$

All state and input **constraints are satisfied** in $\mathcal{X}_f$:

$$\mathcal{X}_f \in X, \kappa_f(x_i) \in U \quad \forall x_i \in \mathcal{X}_f$$

- 3 Terminal cost is a continuous **Lyapunov function** s.t.

$$l_f(x_{i+1}) - l_f(x_i) \leq -l(x_i, \kappa_f(x_i)) \quad \forall x_i \in \mathcal{X}_f$$

Theorem 1. Under the previous assumptions, the closed-loop system under the MPC control law $u_0^*(x)$ is asymptotically stable and the set $\mathcal{X}_f$ is positive invariant for

$$x(k+1) = Ax(k) + Bu_0^*(x(k))$$

Same thing for non-linear MPC! (f/g(x,u) instead of A/B)

- Extends to non-linear systems, but compute sets is difficult!

2 Practical MPC

2.1 Steady-state Target Problem

- Reference is achieved by the target state x_s if $z_s = Hx_s = r$

- Target state should be a steady-state, i.e. $x_s = Ax_s + Bu_s$

$$\begin{aligned} x_s &= Ax_s + Bu_s \\ z_s &= Hx_s = r \end{aligned} \iff \begin{bmatrix} I - A & -B \\ H & 0 \end{bmatrix} \begin{bmatrix} x_s \\ u_s \end{bmatrix} = \begin{bmatrix} 0 \\ r \end{bmatrix}$$

$\nexists$ solution $\rightarrow \min (Hx_s - r)^\top Q_s (Hx_s - r)$ (closest x to r)

If $\exists$ multiple feasible $u_s \rightarrow$ compute cheapest: $\min u_s^\top R_s u_s$

$$\min_U |z_N - Hx_s|_{P_z}^2 + \sum_{i=1}^{N-1} |z_i - Hx_s|_{Q_z}^2 + |u_i - u_s|_{R}^2$$

2.2 Offset-free Reference Tracking

Reference Tracking

$$\begin{aligned} \Delta x_{k+1} &= x_{k+1} - x_s \\ &= A\Delta x_k + Bu_k - (Ax_s + Bu_s) \\ &= A\Delta x_k + B\Delta u_k \end{aligned}$$

$$\begin{aligned} G_x x &\leq h_x \Rightarrow G_x \Delta x \leq h_x - G_x x_s \\ G_u u &\leq h_u \Rightarrow G_u \Delta u \leq h_u - G_u u_s \end{aligned}$$

Convergence

Assume feasible target with $x_s \in \mathcal{X}, u_s \in \mathcal{U}$, choose terminal weight $V_f(x)$ and constraint $\mathcal{X}_f$ as in regulation case satisfying

$$V_f(x(k+1)) - V_f(x(k)) \leq -I(x(k), Kx(k))$$

and $(A + BK)x \in \mathcal{X} \quad \forall x \in \mathcal{X}_f$ for both

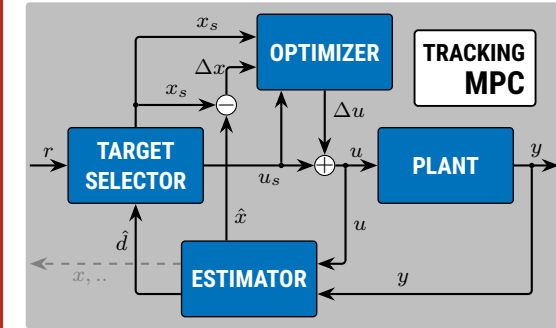
If in addition the target reference x_s, u_s is such that

$$x_s \oplus \mathcal{X}_f \subseteq \mathcal{X}, K\Delta x + u_s \in \mathcal{U}, \quad \forall \Delta x \in \mathcal{X}_f$$

then the closed loop system converges to the target reference.

Proof. Invariance under local control law is inherited from regulation case. Constraint satisfaction is provided by extra conditions and convergence comes from the asymptotic stability of the regulation problem: $\Delta x(k) \rightarrow 0$ for $k \rightarrow \infty$ $\square$

Terminal set use $\mathcal{X}_f^{\text{scaled}} = \alpha \mathcal{X}_f$ (s.t. constraints satisfied)



Disturbance Cancellation

Approach Model the disturbance, use the measurements and model to estimate the state and disturbance and find control inputs that use the disturbance estimate to remove offset.

Augmented Model

$$\begin{aligned} x_{k+1} &= Ax_k + Bu_k + Bd_k \\ y_k &= Cx_k + Cd_k \end{aligned}$$

Constant disturbance $d_{k+1} = d_k$

Observable iff $\begin{bmatrix} A - I & B_d \\ C & C_d \end{bmatrix}$ has full rank (assuming $n_x = n_d$)

Observer For Augmented Model

$$\begin{bmatrix} \hat{x}_{k+1} \\ \hat{d}_{k+1} \end{bmatrix} = \begin{bmatrix} A & B_d \\ 0 & I \end{bmatrix} \begin{bmatrix} \hat{x}_k \\ \hat{d}_k \end{bmatrix} + \begin{bmatrix} B \\ 0 \end{bmatrix} u_k + \begin{bmatrix} L_x \\ L_d \end{bmatrix} (C\hat{x}_k + C_d\hat{d}_k - y_k)$$

Error Dynamics $\Rightarrow$ choose L s.t error dynamics converge to 0

$$\begin{bmatrix} x_{k+1} - \hat{x}_{k+1} \\ d_{k+1} - \hat{d}_{k+1} \end{bmatrix} = \left(\begin{bmatrix} A & B_d \\ 0 & I \end{bmatrix} + \begin{bmatrix} L_x \\ L_d \end{bmatrix} \begin{bmatrix} C & C_d \end{bmatrix} \right) \begin{bmatrix} x_k - \hat{x}_k \\ d_k - \hat{d}_k \end{bmatrix}$$

Lemma 1. Steady-state of an asym. stable observer satisfies:

$$\begin{bmatrix} A - I & B \\ C & 0 \end{bmatrix} \begin{bmatrix} \hat{x}_\infty \\ u_\infty \end{bmatrix} = \begin{bmatrix} -B_d \hat{d}_\infty \\ y_\infty - C_d \hat{d}_\infty \end{bmatrix} \quad (\text{for } n_y = n_d)$$

$\Rightarrow$ Observer output $C\hat{x}_\infty + C_d\hat{d}_\infty$ tracks y_∞ without offset

Reference Tracking with Disturbance Cancellation

Goal Track constant reference: $Hy(k) = z(k) \rightarrow r, k \rightarrow \infty$

$$x_s = Ax_s + Bu_s + Bd\hat{d}_\infty$$

$$z_s = H(Cx_s + C_d\hat{d}_\infty) = r$$

$$\begin{bmatrix} A - I & B \\ HC & 0 \end{bmatrix} \begin{bmatrix} x_s \\ u_s \end{bmatrix} = \begin{bmatrix} 0 \\ r - HC_d\hat{d} \end{bmatrix}$$

Offset-free Tracking - Main Result

Theorem 2. Assuming RHC recursively feasible, $n_d = n_y$, unconstrained for $k \geq j$, and the closed loop system

$$\begin{aligned} x(k+1) &= Ax(k) + B\kappa(\cdot) + B_d d \text{ with } (\cdot) = (\hat{x}, \hat{d}, r) \\ \hat{x}(k+1) &= (A + L_x C)\hat{x}(k) + (B_d + L_x C_d)\hat{d}(k) \\ &\quad + B\kappa(\cdot) - L_x y(k) \\ \hat{d}(k+1) &= L_d C\hat{x}(k) + (\mathbb{I} + L_d C_d)\hat{d}(k) - L_d y(k) \end{aligned}$$

converges, then $z(k) = Hy(k) \rightarrow r$ as $k \rightarrow \infty$

2.3 Soft Constraints

Input constraints are dictated by physical constraints on the actuators and are **usually hard**

State/output constraints arise from practical restrictions on the allowed operating range and are **rarely hard**

Hard state/output constraints always lead to **complications in the controller implementation**

Soft Constrained MPC

$$\min_{u \in \mathcal{U}} \sum_{i=0}^{N-1} x_i^\top Q x_i + u_i^\top R u_i + l_\epsilon(\epsilon_i) + x_N^\top P x_N + l_\epsilon(\epsilon_N) \quad (2)$$

Quadratic penalty	Constraints
$l_\epsilon(\epsilon_i) = \epsilon_i^\top S \epsilon_i$ (e.g $S = Q$)	$x_{i+1} = Ax_i + Bu_i$ $H_x x_i \leq k_x + \epsilon_i$ $H_u u_i \leq k_u$ $\epsilon_i \geq 0$ slack variable
+ linear norm penalty	
$l_\epsilon(\epsilon_i) = v \epsilon_i _1 / \infty$	

Original	$\min_z f(z)$	Softened	$\min_{z, \epsilon} f(z) + l_\epsilon(\epsilon)$
	s.t. $g(z) \leq 0$		s.t. $g(z) \leq \epsilon$ $\epsilon \geq 0$

Requirement on $l_\epsilon(\epsilon)$ If the original problem has a feasible solution z^* , then the softened problem should have the same solution z^* , and $\epsilon = 0$.

Theorem 3 (Exact Penalty Funtcion). $l_\epsilon(\epsilon) = v \cdot \epsilon$ satisfies requirement for any $v > \lambda^* \geq 0$, where λ^* is optimal Lagrange multiplier for original problem

3 Robust MPC

Nominal MPC has some **inherent robustness**, especially with soft constraints, providing Input-to-State Stability (ISS) with respect to small disturbances in an enlarged feasible set. However, it lacks guaranteed robust constraint satisfaction and quantitative stability for larger disturbances. Robust MPC explicitly accounts for known disturbance/model-error bounds to ensure robustness by design.

Uncertain System (Additive Disturbance Case)

$$x(k+1) = Ax(k) + Bu(k) + w(k), \quad w(k) \in \mathcal{W}$$

Inherent Robustness of Nominal MPC

For linear systems with bounded additive disturbances, soft-constrained nominal MPC (with slack variables on state con-

straints and input constraints) is ISS in the soft feasible set $\mathcal{X}_N^{\text{soft}}$ for sufficiently small disturbances.

Key: Optimal cost V_N^* is continuous and serves as an ISS-Lyapunov function for small w .

Limitations: No guarantees for large disturbances; hard constraints may be violated.

Min-Max Robust Formulation

A direct approach to robustness:

$$\begin{aligned} \min_U \max_W \sum_{i=0}^{N-1} \ell(x_i, u_i) + \ell_f(x_N) \\ \text{s.t. } x_{i+1} &= Ax_i + Bu_i + w_i, \\ \forall w_i \in \mathcal{W}, \quad x_i &\in \mathcal{X}, \quad u_i \in \mathcal{U}, \quad x_N \in \mathcal{X}_f \end{aligned}$$

This is computationally expensive (non-convex, scenario tree explodes).

3.1 Robust Open-Loop MPC

Predict open-loop inputs; tighten constraints using disturbance reachable sets assuming no future feedback.

Disturbance reachable set $\mathcal{F}_i = \bigoplus_{j=0}^{i-1} A^j \mathcal{W}$

Robust Open-Loop MPC

$$\begin{aligned} \min_U \sum_{i=0}^{N-1} \ell(x_i, u_i) + \ell_f(x_N) \\ \text{s.t. } x_{i+1} &= Ax_i + Bu_i \\ x_0 &= x(k) \\ x_i &\in \mathcal{X} \ominus \mathcal{F}_i \\ u_i &\in \mathcal{U} \\ x_N &\in \mathcal{X}_f \ominus \mathcal{F}_N \end{aligned}$$

Conservative as it ignores future feedback.

3.2 Closed-Loop Robust MPC (Constraint-Tightening)

Separate control: nominal $z(k+1) = Az(k) + Bv(k)$ and error compensation $u_i = v_i + K(x_i - z_i)$, with K stabilizing ($A_K = A + BK$ Schur stable).

Error dynamics: $e_{i+1} = A_K e_i + w_i$, assuming $e_0 = 0$ (since $z_0 = x(k)$).

Reachable sets: $\mathcal{F}_0 = \{0\}$, $\mathcal{F}_{i+1} = A_K \mathcal{F}_i \oplus \mathcal{W}$.

Constraint-Tightening MPC

$$\begin{aligned} \min_{Z, V} \sum_{i=0}^{N-1} \ell(z_i, v_i) + \ell_f(z_N) \\ \text{s.t. } z_{i+1} &= Az_i + Bv_i \\ z_0 &= x(k) \\ z_i &\in \mathcal{X} \ominus \mathcal{F}_i \\ v_i &\in \mathcal{U} \ominus K\mathcal{F}_i(\delta_i), \\ z_N &\in \mathcal{X}_f \ominus \mathcal{F}_N \end{aligned}$$

Control Law

$$u(k) = v_0^*(x) + K(x(k) - z_0^*(x)) = v_0^*(x)$$

Properties Recursive feasibility, robust constraint satisfaction, ISS w.r.t. w on feasible set.

3.3 Rigid Tube MPC

Use minimal RPI set $\mathcal{E} = \mathcal{F}_\infty = \lim_{i \rightarrow \infty} \mathcal{F}_i$ (smallest set invariant under error dynamics).

Assume initial error $e_0 \in \mathcal{E}$, optimize z_0 s.t. $x \in z_0 \oplus \mathcal{E}$.

Tube: $x_i \in z_i \oplus \mathcal{E}$.

Tighten uniformly: $z_i \in \mathcal{X} \ominus \mathcal{E}$, $v_i \in \mathcal{U} \ominus K\mathcal{E}$.

Rigid Tube MPC

$$\text{Cost } J(Z, V) := \sum_{i=0}^{N-1} \ell(z_i, v_i) + \ell_f(z_N)$$

$$\text{Feasible set } \mathcal{Z}(x) := \begin{cases} z_{i+1} &= Az_i + Bv_i \\ z_i &\in \mathcal{X} \ominus \mathcal{E} \\ v_i &\in \mathcal{U} \ominus K\mathcal{E} \\ z_N &\in \mathcal{X}_f \ominus \mathcal{E} \\ x &\in z_0 \oplus \mathcal{E} \end{cases}$$

$$\text{Control Optimization } \mu_{\text{rigid}}(x) := K(x - z_0^*(x)) + v_0^*(x)$$

$$(V^*(x), Z^*(x)) = \operatorname{argmin}_{V, Z} \{J(Z, V) | (Z, V) \in \mathcal{Z}(x)\}$$

Properties Robust invariance of feasible set; state converges to $\mathcal{E}$.

3.4 Homothetic Tube MPC

For state/input-dependent uncertainty: $w(k) \in \mathcal{W}(x(k), u(k))$, e.g., $w = \Delta Ax + \Delta Bu + d$, $(\Delta A, \Delta B) \in \Omega$, $d \in \mathcal{D}$.

Use scaled tubes: $\mathcal{F}_i = \{e \mid V(e) \leq \delta_i\}$, where $V(e) = \|e\|_M$ is a Lyapunov function (designed offline with contraction $\rho < 1$).

Predict $\delta = (\delta_0, \dots, \delta_N)$ online.

Tube propagation: $\delta_{i+1} \geq \rho \delta_i + \bar{w}(z_i, v_i)$, $\bar{w}(z, v) = \max_{w \in \mathcal{W}(z, v)} \|w\|_M$.

Homothetic Tube MPC

$$\begin{aligned} \min_{Z, V, \delta} \sum_{i=0}^{N-1} \ell(z_i, v_i) + \ell_f(z_N) \\ \text{s.t. } z_{i+1} &= Az_i + Bv_i, \\ \delta_{i+1} &\geq \rho \delta_i + \bar{w}(z_i, v_i), \\ z_i &\in \mathcal{X} \ominus \mathcal{F}_i(\delta_i), \\ v_i &\in \mathcal{U} \ominus K\mathcal{F}_i(\delta_i), \\ (z_N, \delta_N) &\in \mathcal{X}_f, \\ V(x - z_0) &\leq \delta_0. \end{aligned}$$

$$\text{Control Law } u(k) = v_0^*(x) + K(x(k) - z_0^*(x))$$

Offline: Solve LMI for M, K ensuring V decrease under uncertainty.

Constraints often SOCP via Cauchy-Schwarz.

Properties: Less conservative; quantitative convergence to small RPI set $\{x \mid \|x\|_M \leq \max_{d \in \mathcal{D}} \|d\|_M / (1 - \rho)\}$.

Comparison

Constraint-tightening: Fixed tubes, conservative for dependent uncertainty. Homothetic: Predicts tube sizes, reduces conservatism, higher computation.

Offline Design: LMIs & Schur Complement

To find M and K such that $\tilde{A}_K^T M \tilde{A}_K \leq \rho^2 M$ for all $\tilde{A}_K \in$

vertex(Ω): 1. **Schur Complement:** $\begin{bmatrix} \rho^2 M & \tilde{A}_K^T M \\ M \tilde{A}_K & M \end{bmatrix} \succeq 0$.

Change of Vars: Set $E = M^{-1}$ and $Y = KE$ to get linear matrix inequalities (LMI):

$$\begin{bmatrix} \rho^2 E & (AE + \Delta A_i E + BY)^T \\ AE + \Delta A_i E + BY & E \end{bmatrix} \succeq 0$$

Constraint Tightening (Ellipsoidal)

For a state constraint $a_j^T x \leq b_j$, the tightened version is:

$$a_j^T z \leq b_j - c_{x,j} \delta \text{ with } c_{x,j} = \|E^{1/2} a_j\|_2.$$

3.5 Robust Stability (ISS)

A system is **ISS** in Γ if there exist functions $\beta \in \mathcal{KL}$ and $\gamma \in \mathcal{K}$ such that: $\|x(k)\| \leq \beta(\|x(0)\|, k) + \gamma(\|w\|_\infty)$.

- Soft Constraints** Achieve ISS by relaxing $\mathcal{X}$ with slacks ϵ .

- Tube MPC** Achieves ISS; state x converges to the RPI set $\mathcal{E}$.

- Comparison** Fixed tubes (offline $\mathcal{F}_i$) are simpler; homothetic tubes (online δ_i) are less conservative but require SOCP/LMI solvers.

Implementation Checklist:

- Design K (LQR) and M (Lyapunov) offline via LMIs.
- Pre-compute tightenings $c_{x,j}$, $c_{u,j}$.
- Solve QP (Fixed) or SOCP (Homothetic) online.
- Apply $u = v_0 + K(x - z_0)$.

4 Stochastic MPC

Stochastic MPC extends deterministic MPC to handle uncertainties modeled as random variables, such as additive noise or parameter variations. A core feature is replacing hard constraints with chance constraints to account for uncertainty.

$$\mathbb{P}(x_k \in \mathcal{X}) \geq 1 - \epsilon, \quad \mathbb{P}(u_k \in \mathcal{U}) \geq 1 - \delta,$$

This softens constraints, improving feasibility over robust MPC. Chance constraints can be joint or separate (per state/input or timestep).

Mean Dynamics $\bar{x}_{k+1} = A\bar{x}_k + B\bar{u}_k$ (Deterministic)

Variance Dynamics $\Sigma_{k+1} = (A + BK)\Sigma_k(A + BK)^\top + \Sigma_w$

The variance Σ_k can be precomputed offline as it does not depend on the actual state measurements.

For general distributions or polytopic sets, we use **Probabilistic Reachable Set** $\mathcal{F}_k^p$ such that $\Pr(e(k) \in \mathcal{F}_k^p) \geq p$, where $e_k = x_k - \bar{x}_k$.

The constraint is satisfied if: $\bar{x}_k \in \mathcal{X} \ominus \mathcal{F}_k^p$

For Gaussian noise ($w_i \sim \mathcal{N}(0, \Sigma_w)$), $\mathcal{F}_k^p$ can be an ellipsoid: $\{e \mid e^\top \Sigma_k^{-1} e \leq \chi_{n_x}^{-2}(p)\}$, where $\chi_{n_x}^{-2}$ is the inverse chi-squared quantile.

Stochastic MPC

$$J = \mathbb{E} \left[\|x_N\|_P^2 + \sum_{k=0}^{N-1} \|x_k\|_Q^2 + \|u_k\|_R^2 \right]$$

Using the property $\mathbb{E}[x^\top Q x] = \bar{x}^\top Q \bar{x} + \text{tr}(Q \Sigma_x)$, the cost decomposes into:

Nominal Cost A standard quadratic cost on the mean trajectories $(\bar{x}, \bar{u})$.

Constant Offset A term involving $\sum \text{tr}((Q + K^\top R K) \Sigma_k) + \text{tr}(P \Sigma_N)$, which depends only on the noise statistics and the gain K .

Conclusion For a fixed K , optimizing the expected cost is equivalent to optimizing the nominal cost of the mean system.

System dynamics $x_{i+1} = Ax_i + Bu_i + w_i$ (probabilistic)
Chance constraints $\mathbb{P}(x_i \in \mathcal{X}) \geq 1 - \epsilon, \mathbb{P}(u_i \in \mathcal{U}) \geq 1 - \delta$

Initial condition $x_0 = x_k$
Terminal set $\mathbb{P}(x_N \in \mathcal{X}_f) \geq 1 - \epsilon$ (probabilistically)

Tube-like Feedback Policy (to control spread of trajectories)

$$u(k) = \bar{u}_k + K(x(k) - \bar{x}_k)$$

For tractability, approximations like affine feedback policies $u_i = Kx_i + v_i$ (where v_i is optimized) are used to close the loop in predictions. This contrasts with open-loop predictions, which ignore feedback in the horizon and may lead to conservative results.

Feasibility

Large (unbounded) disturbances can drive the system into regions where the optimization problem becomes infeasible at the next time step.

Asymptotic Average Performance Bound

For a stabilizing gain K satisfying the Lyapunov equation $P = (A + BK)^\top P(A + BK) + (Q + K^\top R K)$, $\lim_{T \rightarrow \infty} \frac{1}{T} \sum \mathbb{E}[l(x, u)] \leq \text{tr}(P \Sigma_w)$.

Scenario Approach

For non-Gaussian uncertainties, sample M disturbance scenarios $w^{(j)}$ and enforce constraints for each. Choose M based on desired violation probability (e.g., via Chernoff bounds). This is computationally intensive but distribution-free.

Stochastic vs. Robust MPC: Stochastic allows probabilistic violations for better performance/feasibility; robust ensures worst-case guarantees but can be conservative.

4.1 Recovery Initialization

If the measured state $x(k)$ is infeasible for the MPC, the solver is initialized with the **predicted mean** from the previous step $(\bar{x}_{1|k-1})$ instead of the actual measurement. This ensures recursive feasibility but sacrifices performance.

4.2 Indirect Feedback SMPC

A more advanced formulation where the nominal state $z(k)$ evolves independently of the measurement $x(k)$ in the constraints, but $x(k)$ enters the **cost function**.

- **Pro** Guarantees recursive feasibility and satisfaction of closed-loop chance constraints.

- **Con** The feedback is "weaker" than direct state-feedback MPC.

5 Nonlinear NMPC

5.1 Nominal NMPC

Nominal nonlinear MPC solves a non-convex optimization problem at each step:

$$\min_{X, U} l_f(x_N) + \sum_{i=0}^{N-1} l(x_i, u_i)$$

s.t. $x_{i+1} = f(x_i, u_i), i = 0, \dots, N-1,$

$x_i \in \mathbb{X}, u_i \in \mathbb{U}, i = 0, \dots, N-1,$

$x_N \in \mathbb{X}_f, x_0 = x(k).$

The control law is $\kappa_N(x) = u_0^*(x)$. Assumptions for recursive feasibility and asymptotic stability include: - Positive definite stage cost $l(x, u) \geq \alpha(\|x\|)$. - Terminal set $\mathbb{X}_f$ and cost l_f ensuring local Lyapunov decrease.

Soft-constrained variants add slack variables ϵ_i to handle infeasibility, e.g., penalizing violations in the cost.

5.2 Optimization

Because NMPC requires solving NLPs in real-time, algorithmic efficiency is critical. Optimality is characterized by KKT conditions (stationarity, primal/dual feasibility, complementarity).

Sequential Quadratic Programming (SQP) The standard method for NMPC. It solves a sequence of Quadratic Programs (QP) that approximate the NLP locally using a quadratic cost and linearized constraints. Hessian approximations (e.g., BFGS updates) are used in practice.

Sequential vs. Simultaneous Methods

Sequential (Single-shooting) Only control inputs are decision variables, states are eliminated via simulation. Results in small, dense QPs.

Simultaneous (Multiple-shooting) Both states and inputs are decision variables. Results in large, sparse QPs. (Preferred in practice for better convergence and handling of unstable dynamics)

Real-Time Iteration (RTI) To meet strict timing constraints, RTI performs only one SQP iteration per sampling time, using warm-starts from the previous step.

5.3 Robust NMPC

In linear systems, the error dynamics $e = x - z$ are independent of the nominal state. In nonlinear systems, the **error dynamics** are **state-dependent**, making robust design much harder.

Incremental Stability

Incremental Lyapunov function $V_\delta(x, z)$ such that:

$$V_\delta(f(x, u), f(z, v)) \leq \rho V_\delta(x, z), \quad \rho < 1.$$

This characterizes how "close" the real trajectory x stays to the nominal trajectory z .

Nonlinear Homothetic Tubes The tube is defined as the level set of the incremental Lyapunov function: $\mathbb{X}_\delta = \{x \mid V_\delta(x, z_i) \leq \delta_i\}$. The tube size δ_i is propagated dynamically:

$$\delta_{i+1} = \bar{\rho} \delta_i + \bar{w}(z_i, v_i)$$

where $\bar{w}$ accounts for the worst-case model mismatch/disturbance. Terminal set must be robustly invariant under incremental stability.

Constraint Tightening Nominal constraints are tightened based on the current tube size δ_i , ensuring that if the nominal state

is in the tightened set, the real state is guaranteed to satisfy original constraints.

5.4 Stochastic NMPC

SNMPC aims for **PRS** rather than worst-case tubes. A set $\mathcal{F}_p$ is a PRS of level p if $\Pr(e(i) \in \mathcal{F}_p) \geq p, \forall i$.

The constraints are tightened using the PRS (e.g., $\bar{x}_k \in \mathbb{X} \ominus \mathcal{F}_p^k$ for chance constraints $\Pr(x(k) \in \mathbb{X}) \geq p$).

The cost function is typically the **expected cost** conditioned on the current state (e.g., $\mathbb{E}[l(x, u) \mid x(k)]$).

Indirect feedback schemes use nominal predictions z_i and ancillary control $v_{i,i}$ with input $u(k) = K(x(k) - z(k)) + v_0^*$, ensuring probabilistic guarantees via PRS.

6 Parameter Estimation Learning Based MPC

The fundamental goal of LBMPC is to improve performance by learning unknown system dynamics $g(x, u)$ online while maintaining the stability and safety guarantees of robust MPC.

6.1 Overview of Methods

The main LBMPC approaches are:

Feature	RP-LBMPC	RA/SM-MPC	GP-MPC
Type	Robust	Robust	Stochastic
Regression	Any	Set-Memb.	BLR/GP
Uncert. Update	✓	✓	[✓]
Guarantees	✓	✓	[Some]

RP-LBMPC: Robust Performance LBMPC; RA-MPC: Robust Adaptive MPC; SM-MPC: Set-Membership MPC; GP-MPC: Gaussian Process MPC.

6.2 Regression Techniques for Model Learning

Before controlling the system, one must quantify the uncertainty.

Parametric vs. Non-parametric Parametric models (like BLR) use a finite set of weights θ . Non-parametric models (like GPs) use the data points themselves.

Robust vs. Stochastic Robust methods (Set-Membership) provide hard bounds. Stochastic methods (BLR, GP) provide probability distributions.

Classification: - Parametric Robust: Parametric Set-Membership. - Non-parametric Robust: Non-parametric Set-Membership (not typically covered). - Parametric Stochastic: Bayesian Linear Regression. - Non-parametric Stochastic: Gaussian Processes.

- Bayesian Linear Regression (BLR)** Assumes $y = \Phi(x)^\top \theta + w$. It updates a Gaussian posterior $p(\theta | \mathcal{D}) = \mathcal{N}(\mu_\theta, \Sigma_\theta)$ using Bayes' rule.
- Gaussian Processes (GP)** A non-parametric method defined by a mean $m(x)$ and a kernel $k(x, x')$. It provides a mean prediction $\hat{f}(x_*)$ and a variance $\text{var}[f(x_*)]$.
- Set-Membership (SM) Estimation** A robust approach. Given a bound on noise $w \in \mathcal{W}$, it computes a **non-falsified set** Δ_i of parameters consistent with the data. The parameter set Ω_i is updated via intersection: $\Omega_i = \Omega_{i-1} \cap \Delta_i$.

6.3 Parameter Estimation Methods

For systems with parametric uncertainty in θ and disturbances/noise w_k (assuming full state measurements, simplifying regression):

$$x_{k+1} = f(x_k, u_k, \theta) + w_k$$

Estimation refines θ using observed data. Divided into offline (pre-computed) and online (real-time) approaches. Links to regression techniques above for uncertainty quantification.

Offline Estimation Uses historical data for initial θ fitting.

Least Squares (LS): Minimizes residual sum:

$$\hat{\theta} = \arg \min_{\theta} \sum_{i=1}^N \|y_i - \phi_i^\top \theta\|^2$$

where ϕ_i is the regressor vector.

Extensions: Recursive LS (RLS) for incremental updates, Bayesian methods for uncertainty quantification.

Online Estimation Adapts θ during operation, enabling learning.

Kalman Filter (KF)-based: Treats θ as augmented state, updating via prediction-correction.

$$\hat{\theta}_{k|k} = \hat{\theta}_{k|k-1} + K_k(y_k - \hat{y}_{k|k-1})$$

where K_k is Kalman gain.

Machine Learning Integration: Gaussian Processes (GP) or Neural Networks (NN) for nonparametric estimation, capturing complex θ dynamics. GP models uncertainty as:

$$f(\cdot) \sim \mathcal{GP}(m(\cdot), k(\cdot, \cdot))$$

with mean m and kernel k .

6.4 Robust Performance LBMPC

Core objective is to **decouple** performance from safety.

The controller uses two models simultaneously:

Nominal Linear Model

$$z_{i+1} = Az_i + Bu_i$$

Used to ensure safety. Constraints are tightened relative to this model using Robust MPC techniques (e.g., tube-based MPC).

Learned Nonlinear Model

$$x_{i+1} = Ax_i + Bu_i + \mathcal{O}_k(x_i, u_i)$$

The *Oracle* $\mathcal{O}_k$ represents the learned estimate of dynamics g . The **MPC cost function** is evaluated using this model to achieve high performance.

Result Even if the learned model is wrong, the nominal model ensures the system remains within the *safe* tube. Guarantees: Robust stability and constraint satisfaction.

6.5 Robust Adaptive MPC (Set-Membership Update)

While the Performance model can be any black box, **Uncertainty Set** $\mathcal{W}$ must be updated carefully to avoid losing feasibility.

Mechanism As new data arrives, the set Ω shrinks.

Fixed Shape Parameter Set To prevent the number of constraints from growing to infinity, the set Ω is over-approximated by a polytope of fixed shape, where only the offsets h are updated via a Linear Program (LP).

Guarantee Because $\Omega_{i+1} \subseteq \Omega_i$, the feasible region of the MPC is non-decreasing ($X_{feas}^{i+1} \supseteq X_{feas}^i$), ensuring **recursive feasibility** and robust stability.

6.6 Stochastic Learning-based MPC (Cautious MPC)

When using GPs, the uncertainty is propagated through the horizon. The MPC problem minimizes expected cost with chance constraints:

$$\min \mathbb{E} \left(l_f(x_N) + \sum_{i=0}^{N-1} l(x_i, u_i) \right)$$

s.t. $x_{i+1} = f(x_i, u_i) + B_d g(x_i, u_i) + B_w w_i$, $g \sim \mathcal{GP}(m_g, \Sigma_g)$, $w_i \sim \mathcal{N}(0, \Sigma_w)$, $\Pr(x_i \in \mathcal{X}) \geq p_x$, $\Pr(u_i \in \mathcal{U}) \geq p_u$.

Uncertainty Propagation Since propagating a Gaussian through a nonlinear GP is hard, approximations are used:

Mean Equivalent Treat GP input as deterministic at its mean.

Taylor Expansion Linearize the GP mean around the current state mean.

Exact Moment Matching Analytically compute mean and variance (possible for RBF kernels).

Cautious MPC The controller is *cautious* because the variance $\Sigma(x)$ appears in the constraints (chance constraints) and the cost function. Guarantees: Probabilistic safety and performance improvement with learning.

7 Safety Filter

Learning-based controllers (like Reinforcement Learning) or human operators are excellent at exploring complex tasks but often lack formal safety guarantees.

The idea of **Safety Filters** is to filter the nominal input u^* and produce a safe input u that minimally deviates from u^* while ensuring the system state remains in a safe set.

7.1 Decoupling

Instead of building safety directly into a complex objective function, we decouple the problem.

1. **Performance** A controller (e.g., a Learning-based Controller, human or AI input,...) focuses on the complex objectives and provides a suggested input $u_L(k)$.
2. **Safety Filter** A verification module that sits between the controller and the system. It verifies $u_L(k)$ and modifies it **minimally** only if required to ensure constraint satisfaction for all future times.

7.2 Safe Set

A central concept is the **Safe Set** S (usually Maximal Control Invariant Set), which ensures states remain feasible indefinitely under **Safe Control Law** $\pi_S(x)$.

7.3 Implementation Types

Switching SF

Binary choice. If u_L leads to a safe state, use it. Otherwise, *fall back* to the safe input Kx . This can lead to *harsh* interventions.

Minimally Invasive SF

Formulated as a projection problem:

$$\min_{u_0} \|u_0 - u_L(k)\| \quad \text{s.t.} \quad f(x(k), u_0) \in S, \quad u_0 \in \mathcal{U}$$

Control Barrier Function (CBF) SF

Introduces a *dampening* constraint to prevent the system from rushing toward the boundary of the safe set. It uses a parameter $\gamma \in (0, 1]$ to reduce the *remaining budget* until the boundary:

$$V(f(x, u_0)) - V(x) \leq \gamma(1 - V(x))$$

where V is a CBF (e.g., derived from a Lyapunov function).

7.4 Model Predictive Safety Filter (MPSF)

While invariance-based filters are often restricted to small ellipsoidal sets (which are conservative), the **Predictive Safety Filter** uses MPC techniques to look further ahead.

Mechanism It plans a safe forward trajectory of N steps that terminates in a safe set $\mathcal{X}_f$ (ensuring recursive feasibility and invariance).

Implicit Safe Set: The set of all states for which the MPSF is feasible, $\mathcal{X}_{feas}$, is itself a control invariant set. Crucially: $\mathcal{X}_f \subseteq \mathcal{X}_{feas} \subseteq C$. MPSF allows exploring a much larger state space than $\mathcal{X}_f$.

The Formulation

$$\min_{u_0, \dots, u_{N-1}} \|u_0 - u_L(k)\|$$

$$\text{s.t. } x_{i+1} = f(x_i, u_i), \quad (x_i, u_i) \in \mathcal{X} \times \mathcal{U}, \quad x_N \in \mathcal{X}_f$$

Benefit Any state that is *feasible* for this optimization is considered safe. This implicitly defines a much larger safe set $\mathcal{X}_{feas}$ than a simple ellipsoidal S .

Robustness and Performance

Uncertainty In real-world applications (like quadrotor control or autonomous racing), safety depends on a perfect model. To handle disturbances, the MPSF can be extended using **Robust Tube MPC** (e.g., for bounded disturbances) or **Stochastic MPC** techniques (e.g., probabilistic constraint tightening).

Softening the Intervention To prevent the filter from *fighting* the learner, a barrier term can be added to the learner's own cost function to make it naturally prefer safe regions:

$$C_s(x, u) = C(x, u) - \gamma \log(1 - V(x))$$

Advantages and Limitations

Invariance-based SFs scale to high dimensions and are computationally cheap but conservative. MPSFs reduce conservatism (larger safe sets) but require solving an MPC-like optimization online.

8 Iterative Learning Based MPC

8.1 Iterative Learning MPC

In standard MPC, stability and recursive feasibility are guaranteed by terminal sets $\mathcal{X}_f$ and terminal costs L_f . IL-MPC constructs these *ingredients* dynamically from previous trajectories:

Sampled Safe Set (SS^e) The collection of all successful state trajectories recorded up to episode e .

Convex Safe Set (CS^e) The convex hull of SS^e . For linear systems, any convex combination of successful trajectories is also a feasible trajectory. This set is **control invariant**.

Learned Terminal Cost ($P^e(x)$) Defined as a barycentric function over the safe set. It assigns each point the minimum *cost-to-go* based on a convex combination of previous costs.

Guarantees Under linear dynamics, the resulting MPC is recursively feasible, asymptotically stable, and the closed-loop cost is non-increasing across episodes, eventually converging to the infinite-horizon optimal solution.

Extensions The framework can handle infeasible trajectories (via soft constraints), reference tracking, and nonlinear systems (e.g., using value functions for terminal costs and policies for warm-starting optimization).

8.2 Global Tuning Approaches (Bayesian Methods)

Used when the relationship between cost parameters (e.g., Q, R, P) and a high-level performance metric (e.g., lap time) is unknown or non-analytical.

Bayesian Optimization (BO) Uses a **Gaussian Process** as a surrogate model to represent the unknown performance function. An **Acquisition Function** (like Lower Confidence Bound - LCB) suggests the next parameters to test by balancing exploration (uncertainty) and exploitation (predicted performance). Example: Tuning MPCC for racing to minimize lap times.

Safe BO Restricts exploration to parameters that are likely to maintain performance above a minimum safety threshold (e.g., ensuring lap times do not exceed initial + δ).

Contextual BO Adapts the tuning to changing environments or system parameters (e.g., racing different cars) by treating model mismatches as a *context* variable θ .

Bayesian MPC Uses **Thompson Sampling** to sample parameters from a posterior distribution and use them in the MPC, minimizing *cumulative regret*. Provides bounds on regret and unsafe episodes.

8.3 Local Tuning Approaches (Differentiable MPC)

If the performance metric is differentiable, we can use gradient-based methods.

Differentiable MPC Treats the entire MPC solver as a differentiable layer. This allows the computation of gradients of the optimal trajectories $\frac{\partial u^*}{\partial \theta}$ with respect to cost or dynamics parameters.

RL-MPC Uses Reinforcement Learning to update cost parameters θ to minimize long-term cost on the true system, even with model mismatches.

ZipMPC A student-teacher framework. A *Student* (short-horizon MPC) is trained via imitation learning to mimic the behavior of a *Teacher* (computationally expensive long-horizon MPC). Incorporates context (e.g., reference or dynamics parameters) for adaptation.

8.4 Approximate Explicit MPC

Standard MPC requires solving an optimization problem online, which may be too slow for high-frequency applications. Explicit MPC moves the computation offline.

Explicit MPC for Linear Systems

For linear systems with polyhedral constraints, the MPC solution is a **Multiparametric Quadratic Program (mp-QP)**.

The state space is partitioned into polyhedral regions. In each region, the control law is a unique **Piecewise Affine (PWA)** function: $\pi^*(x) = K_i x + k_i$.

Pros/Cons Online evaluation is just a *point-in-polyhedron* search (very fast), but the number of regions grows exponentially with constraints and horizon (high memory usage).

Approximate Explicit MPC for Nonlinear Systems

Nonlinear systems generally lack an analytical explicit solution. Instead, we learn a function approximation $\tilde{\pi}(x) \approx \pi^*(x)$.

Data Collection - Uniform grid sampling: Simple but suffers from curse of dimensionality. - Adaptive sampling: Focuses on high-error regions. - Closed-loop trajectories: Generates realistic data but may require online MPC evaluations.

Regression Techniques Neural Networks (NN) or Kernel Interpolation are used to fit a model to a dataset of offline-solved MPC problems.

Verification and Safety Because approximation errors can break stability/safety, two approaches are used:

1. **Safety Filters** Project the approximate input $\tilde{u}$ onto a control invariant set to ensure constraints are always met. Can handle parameter-varying systems.
2. **Probabilistic Verification** Use the **Hoeffding Bound** to certify the controller's success rate μ with a specific confidence level $(1 - \delta_h)$ based on a finite number of test samples. Deterministic guarantees (e.g., via Lipschitz constants) are possible but hard to verify in practice.

Extensions Include handling non-convex problems (e.g., obstacle avoidance) and integration with safety filters for robustness.

Iterative Learning Based MPC

Sampled Safe Set $SS^e = \{\bigcup_{j=0}^e \bigcup_{k=0}^{\infty} x^j(k)\}$
Convex Safe Set $CS^e = \text{Conv}(SS^e)$. For linear systems, CS^e is **control invariant**.

Barycentric Terminal Cost $P^e(x)$ for $x \in CS^e$:

$$P^e(x) = \min_{\lambda \geq 0} \sum_{j,k} \lambda_k^j V_{k \rightarrow \infty}^j, \quad \text{s.t.} \quad \sum \lambda = 1, \quad \sum \lambda_k^j x^j(k)$$

where $V_{k \rightarrow \infty}^j$ is the observed cost-to-go from state $x^j(k)$.

9 Invariance and Sets

Definition 1 (Positively Invariant Set $\mathcal{O}$). For an autonomous or closed-loop system, the set $\mathcal{O}$ is positively invariant if:

$$x(k) \in \mathcal{O} \Rightarrow x(k+1) \in \mathcal{O}, \quad \forall k \in \{0, 1, \dots\}$$

Definition 2 (Maximal Positively Invariant Set $\mathcal{O}_\infty$). A set that contains all $\mathcal{O}$ is the maximal positively invariant set $\mathcal{O}_\infty \subset \mathcal{X}$

Definition 3 (Pre-Sets). The set of states that in the dynamic system $x(k+1) = g(x(k))$ in one time step evolves into the target set S is the **pre-set** of $S \Rightarrow \text{pre}(S) := \{x \mid g(x) \in S\}$

Lemma 2. **Invariant Sets from Lyapunov Functions**

If $V : \mathbb{R}^n \rightarrow \mathbb{R}$ is a Lyapunov function for $x(k+1) = g(x(k))$, then $Y := \{x \mid V(x) \leq \alpha\}$ is an invariant set for all $\alpha \geq 0$

Proof. Lyapunov property $V(g(x)) - V(x) < 0$ implies that once $V(x(k)) \leq \alpha$, $V(x(j)) < \alpha, \forall j \geq k \rightarrow$ Invariance $\square$

Example System $x(k+1) = Ax(k), A^\top P A - P \prec 0 \prec P$ and resulting Lyapunov function $V(x(k)) = x(k)^\top P x(k)$

Goal Find the largest α s.t the invariant set $Y_\alpha \in \mathcal{X}$

$$Y_\alpha := \{x \mid x^\top P x \leq \alpha\} \subset \mathcal{X} := \{x \mid Fx \leq f\}$$

Equivalent to $\max_\alpha \alpha \quad \text{s.t.} \quad h_{Y_\alpha}(F_i) \leq f_i \quad \forall i \in \{1 \dots n\}$

Theorem 4 (Geometric condition for invariance). Set $\mathcal{O}$ is positively invariant set iff $\mathcal{O} \subseteq \text{pre}(\mathcal{O}) \Leftrightarrow \text{pre}(\mathcal{O}) \cap \mathcal{O} = \mathcal{O}$

Proof. **Necessary** if $\mathcal{O} \not\subseteq \text{pre}(\mathcal{O})$, then $\exists \bar{x} \in \mathcal{O}$ s.t $\bar{x} \notin \text{pre}(\mathcal{O}) \rightsquigarrow \bar{x} \in \mathcal{O}, \bar{x} \notin \text{pre}(\mathcal{O})$, thus $\mathcal{O}$ not positively invariant

Sufficient if $\mathcal{O}$ not positive invariant set, then $\exists \bar{x} \in \mathcal{O}$ s.t $g(\bar{x}) \notin \mathcal{O} \rightsquigarrow \bar{x} \in \mathcal{O}, x \notin \text{pre}(\mathcal{O})$ thus $\mathcal{O} \not\subseteq \text{pre}(\mathcal{O}) \quad \square$

9.1 Control Invariance

Definition 4 (Control Invariant Set). $C \subseteq \mathcal{X}$ control invariant if

$$x(k) \in C \Rightarrow \exists u(k) \in \mathcal{U} \text{ s.t } g(x(k), u(k)) \in C \quad \forall k$$

Definition 5 (Maximal Control Invariant Set C_∞). A set that contains all C is the maximal control invariant set $C_\infty \subset \mathcal{X}$

Intuition For all states in C_∞ exists control law s.t constraints are never violated $\rightsquigarrow$ **The best any controller could ever do**

Pre-set $\text{pre}(S) := \{x \mid \exists u \in \mathcal{U} \text{ s.t } g(x, u) \in S\}$

Set C is control invariant iff: $C \subseteq \text{pre}(C) \Leftrightarrow \text{pre}(C) \cap C = C$

Control Law from Control Invariant Set

Control law $\kappa(x(k))$ will **guarantee** that the system with control invariant set $\mathcal{C}$ satisfies constraints **for all time** if

$$x(k+1) = g(x(k), u(k)) \rightarrow g(x, \kappa(x)) \in \mathcal{C} \forall x \in \mathcal{C}$$

We can use this fact to **synthesize** control law κ

$$\kappa(x) := \operatorname{argmin}\{f(x, u) \mid g(x, u) \in \mathcal{C}\}$$

with f as any function (including $f(x, u) = 0$)

Does not ensure that system will converge
Difficult because calculating control invariant sets is hard
MPC implicitly describes $\mathcal{C}$ s.t easy to represent/compute

Often used as a terminal set $\mathcal{X}_f = \mathcal{C}_\infty$ to guarantee recursive feasibility in MPC formulations.

For nonlinear systems, computing $\mathcal{C}_\infty$ is NP-hard; approximations (e.g., ellipsoidal sets) are common. In robust MPC, extend to robust positively invariant (RPI) sets for disturbance handling.

9.2 Robust Invariance

Definition 6 (Robust Positive Invariant Set $\mathcal{O}^\mathcal{W}$). For the autonomous system $x(k+1) = f(x(k), w(k))$, the set $\mathcal{O}^\mathcal{W}$ is robust positive invariant if:

$$x \in \mathcal{O}^\mathcal{W} \Rightarrow f(x, w) \in \mathcal{O}^\mathcal{W}, \quad \forall w \in \mathcal{W}$$

Given set Ω and dynamic system $x(k+1) = f(x(k), w(k))$,

$$\operatorname{pre}^\mathcal{W}(\Omega) := \{x \mid f(x, w) \in \Omega \forall w \in \mathcal{W}\}$$

Definition 7 (Robust Pre-Sets). The set of states that in the dynamic system $x(k+1) = g(x(k), w(k))$ for all disturbance $w \in \mathcal{W}$ in one time step evolves into the target set Ω is the **pre-set** of $\Omega \Rightarrow \operatorname{pre}^\mathcal{W}(\Omega) := \{x \mid g(x, w) \in \Omega \forall w \in \mathcal{W}\}$

Theorem 5 (Geometric condition for robust invariance). Set $\mathcal{O}^\mathcal{W}$ is robust positive invariant iff $\mathcal{O}^\mathcal{W} \subseteq \operatorname{pre}^\mathcal{W}(\mathcal{O}^\mathcal{W})$

Minimum Robust Invariant Set

$$\mathcal{F}_\infty = \bigoplus_{j=0}^{\infty} A_K^j \mathcal{W}, \mathcal{F}_0 := \{0\} \Rightarrow \mathcal{F}_n = \mathcal{F}_{n+1} = \mathcal{F}_\infty$$

9.3 Computing Invariant Sets and Pre-sets

System for Pre-Set Computation

$$\begin{aligned} x(k+1) &= Ax(k) + Bu(k) \\ u(k) \in \mathcal{U} &:= \{u \mid Gu \leq g\} \\ S &:= \{x \mid Fx \leq f\} \end{aligned}$$

Invariant Pre-Set

$$\begin{aligned} \operatorname{pre}(S) &:= \{x \mid Ax \in S\} \\ &= \{x \mid FFAx \leq f\} \end{aligned}$$

Control Invariant Pre-Set

$$\begin{aligned} \operatorname{pre}(S) &:= \{x \mid \exists u \in \mathcal{U}, Ax + Bu \in S\} \\ &= \{x \mid \exists u \in \mathcal{U}, FFAx + FBU \leq f\} \\ &= \left\{x \mid \exists u \in \mathcal{U}, \begin{bmatrix} FA & FB \\ 0 & G \end{bmatrix} \begin{bmatrix} x \\ u \end{bmatrix} \leq \begin{bmatrix} f \\ g \end{bmatrix}\right\} \end{aligned}$$

This is a **projection** operation

System for Robust Pre-Set Computation

$$\begin{aligned} x(k+1) &= Ax(k) + w(k) \\ \Omega &:= \{x \mid Fx \leq f\} \end{aligned}$$

Robust Invariant Pre-Set

$$\begin{aligned} \operatorname{pre}^\mathcal{W}(\Omega) &:= \{x \mid FFAx + Fw \leq f\} \\ &= \{x \mid FFAx \leq f - \max_{w \in \mathcal{W}} Fw\} \\ &= \{x \mid FFAx \leq f - h_{\mathcal{W}^T}(F)\} \end{aligned}$$

where $h_{\mathcal{W}^T}(F)$ is the **support function** $\sigma_{\mathcal{C}}(a) = \sup_{x \in \mathcal{C}} a^T x$

9.4 Probabilistic Reachable Sets (PRS)

For a target probability p ($\Pr(x \in \mathcal{R}_k) \geq p$), a PRS can be constructed using the incremental Lyapunov function and Markov's inequality:

$$\mathcal{R}_k = \left\{x \mid V_\delta(x, z(k)) \leq \frac{1 - \rho^k}{1 - \rho} \bar{w}\right\}$$

10 Optimization

Mathematical Optimization Problem

Decision variable $x \in \mathbb{R}^n$
Objective function $f : \operatorname{dom}(f) \rightarrow \mathbb{R}$
Inequality constraints g_i ($i \in \#\text{constraints}$)
Equality constraints h_i ($i \in \#\text{constraints}$)
Feasible set $\mathcal{X} := \{x \mid g(x) \leq 0, h(x) = 0\}$

minimize $f(x)$
subject to:
 $g_i(x) \leq 0$
 $h_i(x) = 0$

Feasible point $x \in \operatorname{dom}(f)$ with $g_i(x) \leq 0, h_i(x) = 0$
Strictly feasible point x with strict inequality $g_i(x) < 0$
Optimal value f^* (or p^*) = $\inf\{f(x) \mid g_i(x) \leq 0, h_j = 0\}$
 $f^* = +\infty$: OP infeasible, $f^* = -\infty$: OP unbound below
Optimizer set: $\operatorname{argmin}_{x \in \mathcal{X}} f(x) := \{x \in \mathcal{X} \mid f(x) = f^*\}$

x^* is a **Global Minimum** if $f(x^*) \leq f(x)$
 x^* is a **Local Minimum** if $\exists \epsilon > 0$ s.t. $f(x^*) \leq f(x)$
 $\forall x \in \mathcal{X} \cap B_\epsilon(x^*)$, open ball with center x^* and radius ϵ

10.1 Convex Sets

Definition 8 (Convex Set). Set $\mathcal{C}$ is convex if and only if

$$\theta x + (1 - \theta)y \in \mathcal{C}, \quad \forall x, y \in \mathcal{C}, \quad \forall \theta \in [0, 1]$$

Intersection $\mathcal{C}_1, \mathcal{C}_2$ cv $\Rightarrow \mathcal{C}_1 \cap \mathcal{C}_2$ convex (**cv**)

Image under affine map $\mathcal{C} \subseteq \mathbb{R}^n$ cv $\Rightarrow \{Ax + b \mid x \in \mathcal{C}\}$ cv

Inverse loaM $\mathcal{C} \subseteq \mathbb{R}^m$ cv $\Rightarrow \{x \in \mathbb{R}^n \mid Ax + b \in \mathcal{C}\}$ cv

Definition 9 (Hyperplanes). $\{x \in \mathbb{R}^n \mid a^T x = b\}$

Definition 10 (Halfspaces). $\{x \in \mathbb{R}^n \mid a^T x \leq b\}$ can be **open** (strict inequality) or **closed** (non-strict inequality)

Definition 11 (Polyhedra). intersection of **finite** number of closed halfspaces: polyhedra $\{x \in \mathbb{R}^n \mid A^{q \times n} x \leq b^{q \times 1}\}$

Definition 12 (Polytope). is a **bounded** polyhedron.

Definition 13 (Convex hull). for $\{v_1, \dots, v_k\} \in \mathbb{R}^d$ is:

$$\operatorname{co}(\{v_1, \dots, v_k\}) := \{x \mid x = \sum_i \lambda_i v_i, \lambda \geq 0, \sum_i \lambda_i = 1\}$$

Definition 14 (Ellipsoid). set: $\{x \mid (x - x_c)^T A^{-1} (x - x_c) \leq 1\}$ where x_c is center of ellipsoid, $A \succ 0$ (i.e. positive definite) (Semi-axis lengths are square roots of eigenvalues of A)

Definition 15 (Norm Ball). $B_r(x) := \{\xi \in \mathbb{R}^n : \|\xi - x\|_p < r\}$ where p defines the l_p norm, $p = \{1/2, \dots, \infty\}$

Theorem 6. Minkowski-Weyl

The following statements are equivalent for $\mathcal{P} \subseteq \mathbb{R}^d$

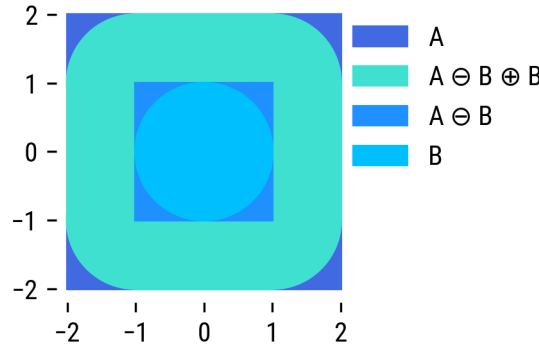
$\mathcal{P}$ is a polytope and there exists A, b s.t $\mathcal{P} = \{x \mid Ax \leq b\}$

$\mathcal{P}$ finitely generated, $\exists$ finite set $\{v_i\}$ s.t $\mathcal{P} = \operatorname{co}(\{v_1, \dots, v_s\})$

Definition 16. Minkowski Sum for $A, B \subset \mathbb{R}^n$

$$A \oplus B := \{x + y \mid x \in A, y \in B\}$$

Example $[a, b] \oplus [c, d] = [a + c, b + d]$



Definition 17. Pontryagin Difference for $A, B \subset \mathbb{R}^n$

$$A \ominus B := \{x \mid x + e \in A, \forall e \in B\}$$

Example $[a, b] \ominus [c, d] = [a - c, b - d]$

10.2 Convex Functions

Definition 18 (Convex Function). $f : \mathcal{C}_{\text{convex}} \rightarrow \mathbb{R}$ is convex iff

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y), \quad \forall x, y \in \mathcal{C}, \quad \forall \theta \in [0, 1]$$

f is strictly convex if this inequality is strict.

Definition 19 (Epigraph). $f : \mathbb{R}^n \rightarrow \mathbb{R} \cup \infty \Leftrightarrow \operatorname{epi}(f)$ is cv set

$$\operatorname{epi}(f) := \{(x, t) \in \mathbb{R}^{n+1} \mid f(x) \leq t\}$$

Check Convexity f is convex if it is composition of simple convex function with convexity preserving operations or if

$$f : \mathbb{R}^n \rightarrow \mathbb{R} \text{ twice differentiable, } \partial^2 f / \partial x^2 \succeq 0 \forall x \in \mathbb{R}^n$$

$g : \mathbb{R} \rightarrow \mathbb{R}$ with $g(t) = f(x + tv)$ convex in $t \forall x, v \in \mathbb{R}^n \rightarrow f$ convex (restriction to a line)

- the point wise maximum of convex functions is convex

- the sum of convex functions is convex

- $f(Ax + b)$ is convex if f is convex

10.3 Optimality Conditions

Lagrange Duality

$$\text{Consider } f^* = \inf_{x \in \mathbb{R}^n} f(x) \text{ s.t. } g(x) \leq 0, h(x) = 0 \quad (3)$$

$$\text{Lagrangian } \mathcal{L}(x, \lambda, \nu) = f(x) + \lambda^T g(x) + \nu^T h(x)$$

$$\text{Dual Function } d(\lambda, \nu) = \inf_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda, \nu)$$

Proposition 1 (Weak Duality). $d(\lambda, \nu) \leq f^*, \forall \lambda \geq 0, \nu \in \mathbb{R}^h$

Definition 20 (Constraint qualification). **Slater's Condition** holds if $\exists$ at least one **strictly feasible point** $\hat{x}$ ($h(\hat{x}) = 0, g(\hat{x}) < 0$)

Proposition 2 (Strong Duality). If Slater's condition holds and OP is convex $\Rightarrow \exists \lambda \geq 0, \nu \in \mathbb{R}^n$ s.t. $d(\lambda, \nu) = f^*$

KKT Conditions (Karush-Kuhn-Tucker)

Theorem 7 (KKT Conditions). If Slater's condition holds and (3) is convex $\rightarrow x^* \in \mathbb{R}^n$ is a minimizer of the primal (3) and $(\lambda^* \geq 0, \nu^*) \in \mathbb{R}^n_g \times \mathbb{R}^n_h$ is a maximizer of the dual $\Leftrightarrow$ is equivalent to the following statements:

KKT-1 (Stationary Lagrangian)	$\nabla_x \mathcal{L}(x^*, \lambda^*, \nu^*) = 0$
KKT-2 (primal feasibility)	$g(x^*) \leq 0, h(x^*) = 0$
KKT-3 (dual feasibility)	$\lambda^*, \nu^* \in \mathbb{R}^n_h \geq 0$
KKT-4 (complementary slackness)	$\lambda^{*T} g(x^*) = 0$ $\nu^{*T} h(x^*) = 0$

In addition we have: $\sup_{\lambda \geq 0, \nu \in \mathbb{R}^n_h} q(\lambda, \nu) = \inf_{x \in \mathcal{C}} f(x)$

Remark Without Slater, KKT-1-4 still implies x^* minimizes (3) and λ, ν maximizes dual, but the converse is no longer true. There can be primal-minimizer/dual-maximizer not satisfy KKT.

10.4 Convex Optimization Problems

Theorem 8. For a convex optimization problem, **any** locally optimal solution is globally optimal (local optima are global optima).

Linear Programming minimize $c^T x$ s.t. $Ax - b \geq 0, x \geq 0$

Step 1: $\mathcal{L}(x, \lambda_1, \lambda_2) = c^T x - \lambda_1^T (Ax - b) - \lambda_2^T x, \lambda_i \geq 0$

Step 2: $\inf_{x \in \mathbb{R}^n} \mathcal{L} = \lambda_1^T b$, if $c - A^T \lambda_1 - \lambda_2 = 0$, else $-\infty$

Step 3: Dual, maximize $b^T \lambda$ s.t. $c - A^T \lambda \geq 0, \lambda \geq 0$ (again LP)

Proposition 3. The optimal solution of a linear program (if it exists) lies always on the boundary of the feasible set and there exists an optimal solution that is a vertex of the feasible set.

Quadratic Programming convex if $P = P^T$ positive semi-definite minimize $\frac{1}{2} x^T P x + q^T x$ s.t. $Gx \leq h, Ax = b$

Second-Order Cone Program

$$\text{minimize } f^T x \text{ s.t. } |A_i x + b| \leq c_i^T x + d_i, Fx = g$$

Second-order cone $C_{n+1} = \{(x, t) \mid x \in \mathbb{R}^n, t \in \mathbb{R}, |x| \leq t\}$

$$|A_i x + b| \leq c_i^T x + d_i \Leftrightarrow (A_i x + b, c_i^T x + d_i) \in C_{n+1}$$

Semi-Definite Programming with symmetric F_i, X, A_i

$$\text{minimize } c^T x \text{ s.t. } \sum_{i=1}^n x_i F_i + G \leq 0, Ax = b$$

CFTOC Optimization

CFTOC problem is **multiparametric quadratic program** **mp-QP**

Two options to solve it:

- Iterative optimization methods

- Explicit solution

Let $I := 1, \dots, m$ be the set of constraint indices.

Definition 21 (Active Set). $A(x)$ and it's complement $NA(x)$

$$\begin{aligned} A(x) &:= \{j \in I : G_j z^*(x) - S_j x = w_j\} \\ NA(x) &:= \{j \in I : G_j z^*(x) - S_j x < w_j\} \end{aligned}$$

Definition 22 (Critical Region). CR_A is set of parameters x for which set $A \subseteq I$ of constraints i active at the optimum. For given $\bar{x} \in \mathcal{K}^*$ let $(A, NA) := (A(\bar{x}), NA(\bar{x}))$. Then

$$CR_A := \{x \in \mathcal{K}^* : A(x) = A\} \quad (\text{states share active set})$$

Online evaluation: Point location

Sequential search

Logarithmic search

Large Scale Convex Optimization

Definition 23 (Lipschitz continuity). $q : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is Lipschitz with constant L if: $|q(x) - q(y)| \leq L|x - y| \forall x, y \in \mathbb{R}^m$

f is Lipschitz (Lip) with constant $L \Leftrightarrow |\nabla f(x)|_2 \leq L$

Theorem 9. $C \subseteq \mathbb{R}^n$ non-empty closed (cl) convex set, $y \notin C \rightarrow \exists a \neq 0, b \in \mathbb{R}$ s.t. $a^T x + b < a^T y + b, \forall x \in C$

Definition 24 (smoothness). $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is L -smooth (L -sm) if $\nabla f(x)$ satisfies $|\nabla f(x) - \nabla f(y)| \leq L|x - y| \forall x, y \in \mathbb{R}^n$

Taylor $\rightarrow f(y) \leq f(x) + \nabla f(x)^T (y - x) + \frac{L}{2}|x - y|^2$
Definition 25 (strong convexity). $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is μ -strongly convex (μ -scv) if $f(y) \geq f(x) + \nabla f(x)^T (y - x) + \frac{\mu}{2}|x - y|^2$

How to find μ/L , Spectra of Hessian $\nabla^2 f$, min/max eigenvalue
Gradient Descent

$x_{k+1} = x_k - T \nabla f(x_k)$ for $k = (k_0, \dots, k_N)$ given x_0, T

Line search optimal step $\nu_t^* = \operatorname{argmin}_{\nu \in \mathbb{R}} f(x_t - \nu \nabla f(x_t))$

Definition 26. $\operatorname{prox}_C(x) = \operatorname{argmin}_{y \in C} \frac{1}{2}|x - y|^2$ $C \subset \mathbb{R}^n$

Lemma 3. cl, cv $C \subset \mathbb{R}^n \rightarrow |\operatorname{prox}_C(x) - \operatorname{prox}_C(y)| \leq |x - y|$
 $\leftarrow |\operatorname{prox}_C(x) - \operatorname{prox}_C(y)|^2 \leq (\operatorname{prox}_C(x) - \operatorname{prox}_C(y))^T (x - y)$

Projected Gradient Descent

$x_{k+1} = \operatorname{prox}_C(x_k - T \nabla f(x_k))$, for $x_0, k_{0..N}, T \in (0, 2/L)$

Lyapunov

Stability is a property of an equilibrium point $\bar{x}$ of a system

Definition 27 (Lyapunov Stability). $\bar{x}$ is Lyapunov stable if: $\forall \epsilon > 0 \exists \delta(\epsilon)$ s.t. $|x(0) - \bar{x}|_2 < \delta(\epsilon) \rightarrow |x(k) - \bar{x}|_2 < \epsilon$

Definition 28 (Globally asymptotic stability). If $\bar{x}$ is attractive, i.e., $\lim_{k \rightarrow \infty} ||x(k) - \bar{x}|| = 0, \forall x(0)$ and Lyapunov stable then $\bar{x}$ is globally asymptotically stable.

Definition 29 (Global Lyapunov function). For the equilibrium $\bar{x} = 0$ of a system $x(k + 1) = g(x(k))$, a function V , continuous at the origin, finite and such that $\forall x \in \mathbb{R}^n$:

$$|x| \rightarrow \infty \Rightarrow V(x) \rightarrow \infty$$
$$V(x) = 0 \text{ if } x = 0 \quad \text{else} \quad V(x) > 0$$
$$V(g(x)) - V(x) \leq -\alpha(x)$$

for continuous positive definite $\alpha : \mathbb{R}^n \rightarrow \mathbb{R}$

then $V : \mathbb{R}^n \rightarrow \mathbb{R}$ is called Lyapunov function.

Theorem 10. If a system admits a Lyapunov function $V(x)$, then $\bar{x} = 0$ is globally asymptotically stable.

Theorem 11 (Lyapunov indirect method). System linearized around $\bar{x} = 0$ with resulting matrix A and eigenvalues λ_i .
If $\forall |\lambda_i| < 1$ then the origin is asymptotically stable.
if $\exists |\lambda_i| > 1$ then origin is unstable.
If $\exists |\lambda_i| = 1$ we can't conclude anything about stability.

Discrete-Time Lyapunov equation $A^T P A - P = -Q, \quad Q > 0 \quad (4)$

Theorem 12 (Existence of solution, DT Lyapunov equation). The discrete-time Lyapunov equation has a unique solution $P > 0$ iff the system $x(k + 1) = Ax(k)$ is stable.

11 Introduction to Systems and Controls

Idea Create a model by solving the systems physical equations

$$x(t) = e^{A^c(t-t_0)}x_0 + \int_{t_0}^t e^{A^c(t-\tau)}B^c u(\tau)d\tau$$

(Exact Solution to ODE of a Linear System)

Problem Most physical systems are nonlinear

Trick use First Order Taylor expansion $f(\bar{x}) + \frac{\partial f}{\partial x^\top} \Big|_{\bar{x}} (x - \bar{x})$

11.1 Linearization

Idea Nonlinear system stable enough around an equilibrium

System equations $\dot{x}_s = g(x_s, u_s) = 0, y_s = h(x_s, u_s)$

Find stationary operating point x_s, u_s and plug in derivative:

$$\Delta \dot{x} = \dot{x} - \dot{x}_s \quad A^c = \frac{\partial g}{\partial x^T} \Big|_{x_s} \quad B^c = \frac{\partial g}{\partial u^T} \Big|_{x_s}$$
$$\Delta y = y - y_s \quad C = \frac{\partial h}{\partial x^T} \Big|_{x_s} \quad D = \frac{\partial h}{\partial u^T} \Big|_{x_s}$$

11.2 Discretization

For general nonlinear systems only approximate discretization methods exist, such as Euler, quality depends on sampling time

Approximation $\dot{x}^c \approx \frac{x^c(t + T_s) - x^c(t)}{T_s}$ Notation $x(k) := x^c(t_0 + kT_s)$
 $u(k) := u^c(t_0 + kT_s)$

Exact Discretization of Linear Time-Invariant Models

$$x(t_{k+1}) = \underbrace{e^{A^c T_s}}_{=A} x(t_k) + \underbrace{\int_0^{T_s} e^{A^c(T_s-\tau)} B^c d\tau}_{B=(A^c)^{-1}(A-I)B^c} u(t_k)$$
$$x(k + N) = A^N x(k) + \sum_{i=0}^{N-1} A^i B u(k + N - 1 - i)$$

11.3 Analysis of Discrete-Time LTI Systems

Controllable if $\operatorname{rank}(C) = n, C = [B \quad \dots \quad A^{n-1}B]$

$\forall (x(0), x^*) \exists$ finite time N with inputs $\mathcal{U}$, s.t. $x(N) = x^*$

Stabilizable iff all uncontrollable modes stable

Observable if $\operatorname{rank}(O) = n, [C^\top \quad \dots \quad (CA^{n-1})^\top]^\top$

$\forall x(0) \exists$ finite time N , s.t. the measurements

$y(0), \dots, y(N - 1)$ uniquely distinguish initial state $x(0)$

Detectable iff all unobservable modes stable

12 Optimal Control

Discrete-Time Optimal Control Problem

Cost Function $J(x_0, U) = \sum_{i=0}^{N-1} l(x_i, u_i) + l_f(x_N) \quad (5)$

Stage Cost	Constraints
$l_f(x_i, u_i)$	$x_{i+1} = g(x_i, u_i)$
Terminal Cost	$x_0 = x(k)$
$l_f(x_N)$	$h(x_i, u_i) \leq 0 \quad (\text{optional})$

12.1 Unconstrained Finite Horizon Control Problem

Linear Quadratic Optimal Control

Cost Function $J^*(x(0)) := \min_U \sum_{i=0}^{N-1} x_i^\top Q x_i + u_i^\top R u_i + x_N^\top P x_N \quad (6)$

Terminal weight	Constraints
$P \succeq 0$ symmetric	No input or state constraints!
State weight	$x(k + 1) = Ax_k + Bu_k$
$Q \succeq 0$ symmetric	
Input weight	Only dynamics matter.
$R \succ 0$ symmetric	

Recursive Approach

use dynamic programming to solve problem backwards from N

$$J_j^*(x(j)) := \min_{U_{j \rightarrow N}} x_N^\top P x_N + \sum_{i=j}^{N-1} x_i^\top Q x_i + u_i^\top R u_i$$

From Principle Of Optimality

Optimal Cost-To-Go

$$J_j^*(x_j) = \min_{u_j} I(x_i, u_i) + J_{j+1}^*(x_{j+1}) \quad J_i^*(x_i) = x_i^\top P_i x_i$$

Optimal Control Policy use Riccati

$$u_i^* = F_i x_i = -(B^\top P_{i+1} B + R)^{-1} B^\top P_{i+1} A \cdot x(i)$$

Comparison of Batch and Recursive Approach

Dynamic programming yields feedback policies $u_i^* = F_i x_i$ depending on each x_i . Batch optimization returns sequence $U^*(x(0))$ of numeric values depending only on $x(0)$,

12.2 Infinite Horizon Control Problem

Linear Quadratic Regulator

Cost Function $J_\infty^*(x(k)) = \min_{u(\cdot)} \sum_{i=0}^\infty x_i^\top Q x_i + u_i^\top R u_i \quad (7)$

Constraints	$x_{i+1} = Ax_i + Bu_i$
	$x_0 = x(k)$
Optimal Input	$u^*(k) = F_\infty x(k) \quad (\text{use Riccati})$

$$F_\infty x(k) = -(B^\top P_\infty B + R)^{-1} B^\top P_\infty A x(k)$$

Lemma 4 (Lyapunov function for LQR). If (A, B) is stabilizable and $Q, R \succ 0$ then $J^*(x) = x^\top P_\infty x$ is a Lyapunov function for the system $x^+ = (A + BF_\infty)x$ where F_∞ is the constant feedback matrix and $P_\infty \succ 0$ solves the Riccati equation..

Choice of P

- 1 Match the infinite solution $P_N = P_\infty$
- 2 Use solution of the Lyapunov Equation (4) if the system is asymptotically stable (otherwise P not positive definite), assumes no control needed after end of horizon.
- 3 Set $P = 0$ und use instead constraint $x_{i+N} = 0$

12.3 Constrained Finite Time Optimal Control Problem Quadratic Cost CFTOC

$$J(x(k)) = x_N^\top P x_N + \sum_{i=0}^{N-1} x_i^\top Q x_i + u_i^\top R u_i$$

CFTOC

Cost Function equal to Linear Quadratic Optimal Control (6)
Constraints as in (6) + $x_N \in \mathcal{X}_f, x_{i < N} \in \mathcal{X}, u_i \in \mathcal{U}$

Goal: Transform into QP

$$\min_{z \in \mathbb{R}^n} \frac{1}{2} z^\top H z + q^\top z + r \quad \text{s.t. } G z \leq h, \quad A z = b$$